

25_ACE + Multi-Agent Termination Study

Automated architectural intelligence pipeline + first systematic study of multi-agent LLM team termination dynamics. End-to-end ownership: AI, geometry, 3D visualization, multi-agent orchestration, deployment.

About

Profile

DongHyeon Kim — full-stack AI engineer working at the intersection of [architectural law](#), [3D computational geometry](#), and [multi-agent orchestration](#).

Built **25_ACE** end-to-end: a production AI pipeline that turns a single Korean address into a fully-resolved 3D building envelope, traversing legal regulations, terrain elevation, and NSGA-II mass optimization automatically.

Authored **Multi-Agent Termination Dynamics** (COLM 2026 target) — first systematic 14-pattern × 7-experiment × 2,200-run unified study answering *"when should multi-agent LLM teams stop?"*

Core Skills

AI / ML

Multi-agent (AutoGen), RAG, embeddings (3072d), G-Eval, NSGA-II, RF prediction

Backend

Django · FastAPI · Python 3.13 · Neo4j · CF Workers · Railway · Oracle Cloud

Frontend / 3D

React 19 · TypeScript · Vite · Tailwind v4 · Cesium 3D · Shapely + pyproj

Research

Experimental design · Kruskal-Wallis · Cohen's d · 5-fold CV · cross-model validation

NUMBERS AT A GLANCE

31K

NEO4J LEGAL NODES

178+

AUTOMATED TESTS

11m

URBAN ACCURACY

14

AGENT TOPOLOGIES

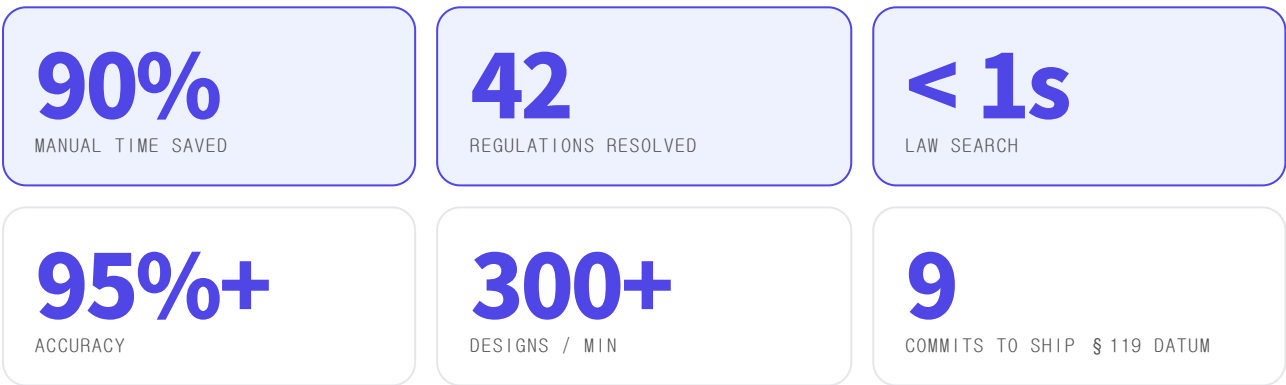
2.2K

RESEARCH RUNS

25_ACE

One address. Full 3D building feasibility. Automatic.

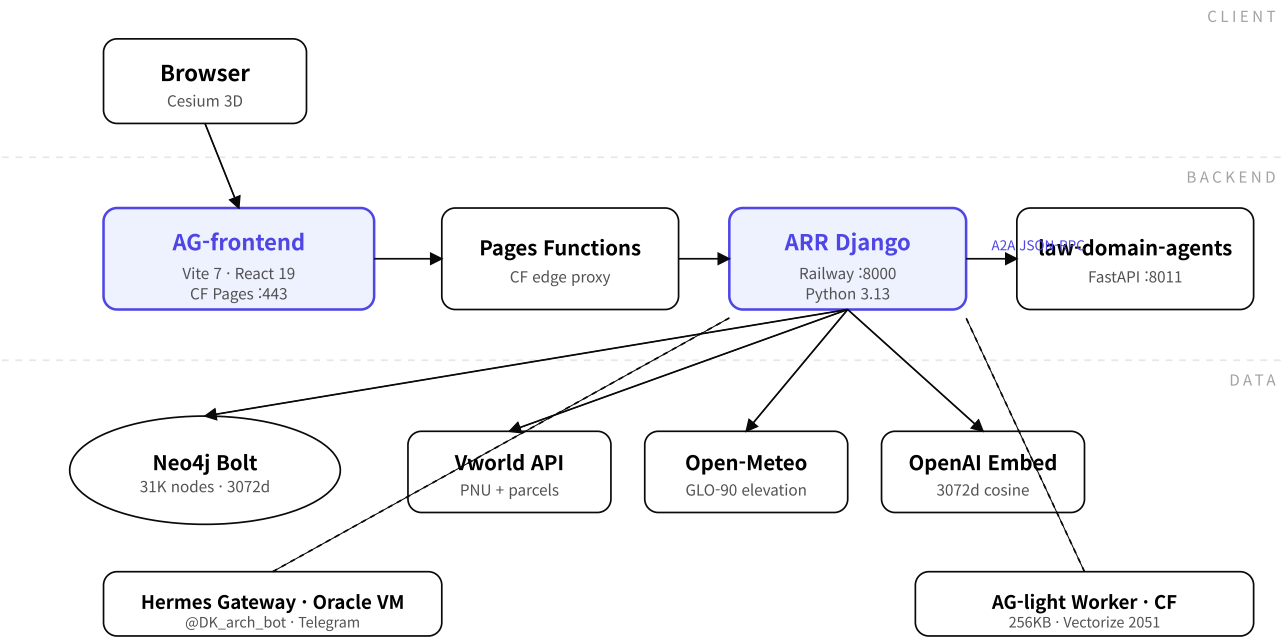
Korean architectural feasibility traditionally requires weeks of manual legal cross-referencing, terrain surveys, and iterative sketching. 25_ACE compresses the entire pipeline into a deterministic chain reaction triggered by a single address string.



PIPELINE AT A GLANCE

Address → Vworld geocoding → PNU 19-digit → Neo4j 7-stage hybrid search → § 119 datum → § 86 envelope
→ NSGA-II Pareto → Cesium 3D

System Architecture

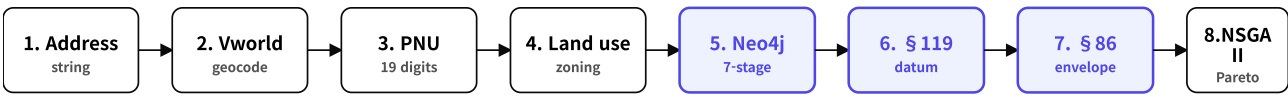


Frontend tier — single-page Vite/React 19 app, served via Cloudflare Pages, proxies API calls through Pages Functions to backend tier.

Backend tier — Django on Railway is the stateful core; law-domain-agents (FastAPI) handles 7-stage hybrid search; communication via A2A JSON-RPC 2.0.

Data tier — Neo4j (legal graph, 31K nodes), Vworld API (geocoding), Open-Meteo (DEM elevation), OpenAI (3072d embeddings).

Address → 3D Mass · 8 Steps



STEP	TECH	OUTPUT	LATENCY
1. Address	user input	"강남구 역삼동 677"	–
2. Vworld geocoding	api.vworld.kr/req/address	WGS84 + refined.level4LC	~120ms
3. PNU extraction	level4LC → 19 digit	1168010100106770003	~1ms
4. Land use lookup	Vworld DataAPI	zoning + parcel area	~200ms
5. Law search	Neo4j 7-stage hybrid	42 regulations + 170 articles	~800ms
6. Datum dispatcher	§ 119 6-case + Open-Meteo	Z-axis reference plane H	~300ms
7. Envelope generation	§ 86 slope 2:1 (LOCKED SPEC)	4-stage stair-step polygon	~50ms
8. Mass optimization	NSGA-II (pop 30, gen 30)	Pareto front + Cesium 3D	~3–8s

End-to-end: <5 sec law analysis · <10 sec full 3D mass · 95%+ accuracy · 178+ tests passing.

Algorithm Stack

Four algorithm families work in concert: [graph search](#) for legal articles, [decision dispatcher](#) for ground level, [multi-objective GA](#) for building mass, and [spatial algorithms](#) for floor plans.

1 • Neo4j 7-Stage Hybrid Search

Korean legal articles as 7-tier graph (LAW → JANG → JEOL → JO → HANG → HO → MOK). Vector + fulltext + relationship boost.

Exact → Fulltext (CJK bigram) → Vector 3072d → Rel boost → RRF → MMR → Domain re-rank

31,126 nodes

95%+ accuracy

2 • § 119 Datum 6-Case Dispatcher

Decision tree for Korean Building Act § 119 datum plane: 6 terrain cases × Open-Meteo GLO-90 elevation.

FLAT (<2m) · SLOPE_LE3M · SLOPE_GT3M · ROAD_FLAT · ROAD_SLOPED · SITE_ABOVE_ROAD

11m urban accuracy

9 commits

3 • § 86 Envelope • LOCKED SPEC

North-side sunlight envelope (slope 2:1, 4-stage stair-step). 3-layer protection: config flag · backend source enum · frontend ternary.

H = 2d · stages every 9m · walls + slanted polygons

3-layer LOCKED

39 datum tests

4 • NSGA-II Multi-Objective GA

Mass optimization: pop 30, gen 30, crossover 0.9, mutation 0.1. 3-objective: FAR vs sunlight vs landscaping.

Pareto front · 10 mass types · 300+ designs/min

10 mass types

Pareto front

10 MASS TYPES · 4 FLOOR PLAN ALGORITHMS

Mass shape variants — NSGA-II searches within each:

Additive (K=5)

Subtractive

Grid 3×3

L-shape

U-shape

Cross

Courtyard

Tower+Podium

H-shape

Radial

Floor plan generators — 4 algorithms compared, 104 tests:

GA + Series Gene

Subdivision

MCTS (UCB tree)

Circle Packing

GA: Pareto multi-obj · Subdivision: fastest deterministic · MCTS: ~300 LOC port · Circle Packing: physics → grid.

Live Deployment + Validation

4-SERVICE PRODUCTION

● AG-frontend

\$0 / mo

ag-frontend-5s3.pages.dev
Cloudflare Pages · Vite 7 build

● ARR Backend

\$5 / mo

arr-backend-production.up.railway.app
Railway · Django 6 + WhiteNoise

● AG-light Worker

\$0 / mo

law-light-api.clickaround8.workers.dev
Cloudflare Workers · 256KB · Vectorize 2,051

● Hermes Gateway

\$22 / mo

158.180.66.165 · @DK_arch_bot
Oracle VM · Telegram bot · Hermes plugin

Total: ~\$28 / mo for production-grade multi-region multi-channel system.

VALIDATION STACK

Unit · 178+ tests

Django test suite. 39 datum-specific (\$ 119 6-case full coverage), 167 land regression, 24 design / mass evaluator.

Integration · Playwright E2E

235 E2E tests across 25 spec files. Full-stack: frontend → backend → DB → 3D render.

31s execution time

Live · 8 PNU production runs

강남 · 성북 · 한남 · 여의도 · 우동 · 평창 · 대관령 등 8개 실측 PNU.

11m urban avg accuracy · 3.7m coastal

9 commits to ship \$ 119 datum stable.

Lean engineering · TDD-driven · zero regressions in 4 months.

Multi-Agent Termination Study

When should multi-agent LLM teams stop?

First systematic study of termination dynamics across 14 coordination topologies. 2,200+ unified runs. Six counterintuitive findings. COLM 2026 target.



14 PATTERNS · 6 CATEGORIES

S	Baseline	Solo
A	Sequential Chain	RR-2RR-3RR-4
B1	Centralized Routing (Star)	SeI-3SeI-4
B2	Decentralized Handoff (Mesh)	Swm-3Swm-4
C	Structured Feedback	RefI-2RefI-3Deb-3Deb-4
D	Composed	Pipe (5)MoA (4)

Six Counterintuitive Findings

Finding 1 • Cost Inversion

3-agent swarm **cheaper than** 2-agent chain.
Swm-3: 3,203 tok vs RR-2: 4,759 tok
U=95, p<0.001

Finding 2 • Debate Quality Decline

Monotonic decline across rounds (Du et al. ICML 2024 contradicted).
3.8 → 3.4 → 3.3 (G-Eval Sonnet 4.5)

Finding 3 • B1 / B2 Split

Centralized vs decentralized scaling diverges.
B1: 1.48 × sub-linear · B2: 3.17 × super-linear
New hierarchy: A ≈ B2 < B1 ≈ C ≪ D

Finding 4 • Keyword Optimal • Negative Result

800-run controlled exp: ΔU adaptive vs keyword TERMINATE.
7/8 patterns: keyword sufficient
Ref1-2 ΔU: +258% overhead · only Swm-4: -70%

Finding 5 • Cost Prediction Model

Random Forest, 5-fold CV.
R² = 0.54 · top feature: agents × max_messages
46% remains task-content (honest boundary)

Finding 6 • Difficulty Dominance

Task difficulty > pattern choice. 5 LLMs · 669 runs.
η² (difficulty) = 0.363 vs η² (pattern) = 0.039
Solo dominates · Ref1-2 helps Haiku@medium d=1.26

ΔU DIAGNOSTIC FRAMEWORK

$$\Delta U(t) = \Delta Q(t) - \lambda \cdot \Delta C(t)$$

All 14 patterns saturate within **1.1–2.4 turns**. λ-insensitive (0.0 → 0.5 yields ≤0.1 turn difference). Use as *diagnostic*, not replacement.

Engineering Stack

React 19 Frontend rendering · concurrent mode	TypeScript Strict type-safe client	Vite 7 Ultra-fast bundler · ESM-first	Tailwind v4 Utility-first CSS · semantic tokens
Django 6 Backend framework · REST + SSE	Python 3.13 Core algorithmic processing	FastAPI law-domain-agents · :8011	Neo4j 5 31K-node legal graph + vector index
Cesium 3D High-precision WebGL globe	Shapely + pyproj Geometry · CRS transforms	OpenAI text-embedding-3-large 3072d	AutoGen Multi-agent (research project)
Cloudflare Pages + Workers + Vectorize	Railway Backend hosting · auto- deploy	Oracle VM Hermes gateway · Telegram bot	Playwright E2E · 235 tests · 31s execution

STATISTICS & VALIDATION

scipy	Kruskal-Wallis	Mann-Whitney U	Cohen's d	5-fold CV	Spearman ρ	G-Eval LLM-judge
sentence-transformers						

5 LLMS TESTED (CROSS-VALIDATED)

Claude Haiku 4.5	Claude Sonnet 4.5	GPT-4o-mini · ρ=0.900	Gemini	GPT-5.4
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Roadmap 2026



WHAT'S NEXT · TECHNICAL

NGII 5m DEM

Self-host elevation tile server.
90m → 5m precision. 18×
accuracy improvement. R2 +
opentopodata.

Hybrid termination

Combine TERMINATE keyword +
 ΔU only for B2 (decentralized
handoff). Validated optimization
for the one topology where
keyword fails.

LangGraph / CrewAI port

Cross-framework validation of
14-pattern findings. Test whether
topology effects generalize
beyond AutoGen.

Let's build the next standard together.

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LIVE DEMO

arr-frontend.pages.dev

@DK_arch_bot · Telegram